Kabir Kanha Arora
1710110165

Instructions:

- There are 03 questions in this assignment.
- **Assignment submitted after due date and time will not be evaluated and a score of zero will be awarded for this assignment.**
- Upload a **pdf version** of the document.
- Materials copied from the Internet or otherwise will attract penalty as per the course policy.
- Make proper citations and attach a reference section.

Due Date: 5:55 pm, January 28, 2019

Submitting this Assignment:

You will submit (upload) this assignment on Blackboard. Email submissions will not be accepted.

- Write your answer after the given question in this document.
  Name this document as **A1_OS2019_John_Doe.pdf** in case your name is John Doe [You are required to strictly follow the naming convention.]

Grading: This assignment has 5 points (with **weightage of 5%** in your overall 100 points).

Questions

1. Compare and contrast Android, iOS, Windows Mobile OS and Blackberry OS based on the following parameters: (800-1000 words) **(2)**
   a) OS services and features
   b) Design challenges - policies and mechanisms

2. Compare microkernels with monolithic OS architecture. (800-1000 words) **(2)**
   Note: Cover all parameters given in the papers

3. Differentiate between reliability of an OS and security of an OS. List the reliability and security issues in an OS. (400-500 words) **(1)**

Answers:

A1

ANDROID

1.

Android is an open source working OS for cell phones created by Google and the Open Handset Alliance. The OS engineering comprises of:

• an adjusted Linux Kernel

• open source Libraries coded in C and C++

• the Android Runtime, which considers center libraries that transfers the most center elements of Java. As virtual machine it utilizes Dalvin, which empowers to execute Java applications.

• an Application OS, which transfers administrations and libraries coded in Java for the application advancement

• and the Applications, which work on it

Android is a stage and furthermore a working OS for cell phones and different gadgets. Android OS is structured particularly for contact screen cell phones yet in addition for (TV Android), vehicles (Android Auto) and watches (Android Wear). It tends to be said that there are numerous equipment items on which an Android OS is running. This OS can change a cell phone into a PC with the span of a pocket number cruncher.

2.

Since adaptation 2.2 Android offers a Device Administration API, that gives gadget organization and control highlights at the OS level. With this API, engineers can compose gadget the executives applications that upholds approaches. These approaches could be hard-coded into the application, or got progressively from an outsider server. The API bolsters following local arrangements, which are divided into required stage variant for help.

The Android OS is an open source programming, which implies that any client can convey enhancements to the working OS, in this way one may profit from Google engineers' know-how, as well as from that of outsider designers. Google opened the whole source code (counting the system and communication bolster) so makers are allowed to include augmentations without making them accessible to the open source network.

Linux is simply part of this engineering and is put at the most minimal dimension. As on account of the OS from Apple, we don't see the Android OS as a Linux working OS, and in this way most designers don't approach at this dimension.

From the perspective of its designer, the Android OS has a Linux piece, in light of Java (with Eclipse module); it has its own Google virtual machine, called Dalvik, and the Linux highlights are not available to clients.

Android Runtime (ART) is utilized by the versatile Android working OS, which was first present in Android 4.4 KitKat; in Android 5.0, the Dalvik virtual machine is supplanted totally. Workmanship replaces Dalvik virtual machine, which does the transforming of the application bytecode into local guidelines each time the individual application begins (just-intime). Craftsmanship presents another technique called early (AOT), arrangement, which is performed at introducing the application.

WINDOWS MOBILE OS

1.

Windows Phone is a successor of the working OS Windows Mobile of the product engineer Microsoft. By correlation with the other examined versatile working OSs, the piece of the pie is low with just 4,2%. Be that as it may, as in the beforehand part referenced can the participation among Nokia and Microsoft be appraised as a high potential for a piece of the pie development.

Windows Phone utilizes advances and instruments, which are additionally utilized in the station based application improvement, similar to the advancement condition Visual Studio and the OSs Silverlight, XNA and .NET Compact. Besides Windows Phone considers a total combination with the Microsoft Services Windows Live, Zune, Xbox Live and Bing. For sandboxing Windows Phone utilizes a similar model like Android and iOS.

2.

Besides outsider applications must be marked with an official accreditation, similar to iOS Applications, The Windows Phone working OS was essentially intended for the client and not the undertaking market and gives no local arrangement settings, similar to Android or iOS. Be that as it may, it offers help for ActiveSync arrangements, which are strategy settings utilized by a Microsoft Exchange Server, likewise a Microsoft item. The Windows Phone successor Windows Mobile bolstered all Exchange ActiveSync strategies, whereby Windows Phone underpins just some essential arrangements like:

- Password Required
- Minimum Password Length
- Idle Timeout Frequency Value
- Allow Simple Password
- Password Expiration

• Password History

The format Windows Phone is likewise an OS intended for brilliant cell phones. This OS was produced by Microsoft. Windows Phone OS is the successor of Windows Mobile stage, which depended on Windows CE portion, beginning with the working OS Pocket PC 2000. Windows Phone 7 OS was declared on February 15, 2010 and was discharged on November 8, 2010. This OS was gone before by Windows Mobile 6.x variant. In sequential request, the Windows Phone adaptations are: Windows Phone 7, Windows Phone 7.5, Windows Phone 7.8, Windows Phone 8 (GDR1, CDR2, GDR3), Windows Phone 8.1 (GDR1, CDR2) and Windows 10 (versatile).

Similarly as with the Android OS, Windows Phone OS is anticipated for different equipment stages and chipsets, so that application designers must have the capacity to "gather" progressively. Microsoft utilized C# dialect as the primary advancement dialect and everything is assembled on the CLR, which is its own virtual machine, like the Dalvik virtual machine of the Android OS.

A component of curiosity is spoken to by the way that notices can be synchronized among gadgets. Another preferred standpoint is the enhanced interface and its application, just as its utilization with one hand. The principle analysis conveyed to Windows Phone OS is the nonappearance of applications contrasted with iOS and Android working OSs. As application stage introduced on cell phones, Windows Store stage is on the third place, behind the stages from Android OS and iOS OS.

iOS

1.

The iOS OS (initially called iPhone OS) is the portable working OS created by Apple Inc. for cell phones produced by Apple. Initially this working OS was created for iPhone, and later it was reached out to be actualized on other Apple gadgets, for example, iPod contact (MP3 player), iPad, Apple TV (set-top box sort of gadget). These gadgets are considerably more costly, yet additionally steady when contrasted with different gadgets and OSs created by contending makers. iOS is a Unix-like working OS, which contained, since its first form, a few components of Mac OS X working OS (Unix-like working OS from Apple Inc.). One of the principle preferences of iOS OS is that Apple permits the refreshing of the working OS for more established tablets, yet this help is declining. Apple gives updates to iOS working OS through over-the-air (OTA) interface. For item strategy reasons, iOS OS does not bolster the Flash media use of Adobe American organization.

The exclusive portable working OS iOS is utilized for Apple gadgets like the iPhone and is a further improvement of the working OS Mac OSX. Its piece of the overall industry became persistently in the course of the most recent year to 15,7%.

2.

The OS design is indistinguishable to the MacOSX engineering and comprises of the accompanying segments:

• Core OS: The piece of the working OS
• Core Services: Fundamental OS administrations, which are subdivided in various structures and dependent on C and Objective C. For instance offers the CFNetwork OS the usefulness to work with known system conventions.
• Media: Considers the abnormal state systems, which are in charge of utilizing realistic , sound and videotechnologies.
• Coca Touch: Includes the UIKIT, which is an Objective-C based system and gives various functionalities, which are vital for the improvement of an iOS Application like the User Interface Management

Like in the Android area referenced, iOS utilizes a comparative sandboxing model. Besides applications must be marked with an issued authentication. This guarantees application have not been controlled and guarantees the runtime to check if an application has not moved toward becoming untrusted since it was last utilized. Uneven Android applications, iOS applications must be marked with an official affirmation.

The iOS gives some strategy insurance that can be conveyed and upheld over the air or locally. This authorization can be controlled and designed by utilizing outsider Mobile Device Management arrangements, wherefore Apple gives a different API.

After an iOS gadget is enrolled with the Mobile Device Management Server, the gadget can be control powerfully. In this manner the Server sends XML setup profiles to the gadget, which empowers a safe use in an undertaking domain. The design profile contains: gadget security strategies and limitations, VPN setup data, Wi-Fi settings, email and schedule records, and confirmation information, that allow to work with the undertaking OS. Preference of iOS is, that the it works just on a couple of Apple gadgets, which implies that there is no high stage variant dissemination like on Android. Not at all like in Android, versatile working OS refreshes impact each io gadget.

Given its usefulness, iOS OS is one of the achievement elements of iPhones on the planet advertise. A contender of iOS is the Android working OS from Google. iOS working OS is not the same as past working OSs (Android and Windows Phone), for the most part in light of the fact that both the working OS and the last items are kept up and created by a similar organization. You can't inspire an iOS OS to keep running on an alternate machine on the grounds that there is no permit to introduce an iOS working OS on equipment that has not been made by Apple. Today there are just test systems available for iOS OS. With iOS, there are four dimensions of reflection: the working OS bit, the center administrations level, the media level and the UI.

iOS is not the same as Android and Windows Phone essentially in that both the working OS and the last item are created and constrained by a similar organization, Apple and you can't get an iOS working OS to chip away at an alternate gadget since there is no permit for the establishment of iOS working OS on equipment gadgets that are not delivered by Apple. This is a disservice since this OS isn't versatile on cell phones delivered by different organizations.

BLACKBERRY OS

1.

Blackberry OS is the exclusive versatile working OS, created by the Canadian organization Research in Motion and is utilized for Blackberry gadgets as it were. Rather than the various respected portable working OSs, it is for the most part created for business utilization. Gartner says that it is a standout amongst the most mainstream portable working OS today with 16,0% piece of the overall industry, yet they likewise foresee a diminishing significance later.

BlackBerry OS utilizes a more seasoned model for application sandboxing. It utilizes diverse trust jobs for assignments and applications have full access to the total gadget and information. It is likewise requiring to sign an application through Certificate Authorities (CA) or created (self-marked) endorsement to run code on the gadget. Moreover the mark gives data about the benefits to an application, which is important on the grounds that applications have full access on BlackBerry gadgets, on account of its sandboxing model.

2.

As an exemplary business cell phone producer, Blackberry gives more than 400 strategies to their versatile working OS, which can be utilized to control explicit portable IT approaches in a venture domain. Generally to Android they must be upheld in blend with their BlackBerry Enterprise Server arrangement. The more than 400 strategies given by BlackBerry offer everything that is important to utilize a cell phone safely in an enterprise domain. BlackBerry furnishes an entirely significant approach arrangement with the disservice, that the Blackberry Enterprise Server is important to implement them. This implies for an association, that they need to utilize Research in Motion gadgets and can't bolster a various arrangement with utilizing distinctive cell phones running on various versatile working OSs.

BlackBerryOS does not bolster any Exchange ActiveSync Policies and utilizations his BlackBerry Enterprise Server answer for authorizing the more than 400 approaches on their gadgets. The accompanying passage will portray how the information trade between the cell phone and BlackBerry Server works, which is important to uphold and synchronize the approaches of a system association. The information trade of the BlackBerry Enterprise Solution goes along the solid necessities of government expert and is guaranteed by the NATO. BlackBerry offers two transport encryption alternatives, Advanced Encryption Standard (AES) and Triple Data Encryption Standard (Triple DES), for information trade between the BlackBerry Enterprise Server and the BlackBerry gadget. The private encryption keys,

who are essential for the encryption, are produced in a safe, two-manner credible condition and are relegated to the Blackberry gadget. Every mystery key is put away on the clients secure undertaking account and on the BlackBerry gadget.

BlackBerry applications expect designers to sign and enroll their applications with Research In Motion. This includes insurance by giving a more noteworthy level of control and consistency to the stacking and conduct of utilizations on BlackBerry gadgets. It is additionally requiring to sign an application through Certificate Authorities (CA) or created (self-marked) testament to run code on the gadget. Besides the mark gives data about the benefits to an application, which is important in light of the fact that applications have full access on BlackBerry gadgets in motivation to the customary sandboxing model. Since arrangement requirement is just conceivable by utilizing the BlackBerry Enterprise Server, the strict application marking arrangement has no effect on it.

A2

PARAMETER	Microkernel OS Architecture	Monolithic kernel OS Architecture
Size	Smaller in size. Minux is a microkernel, defined more or less as a very small operating system that provides system calls to manage basic services like handling threads, address spaces, and inter-process communications. A microkernel relegates all other activities to "servers" that exist in user space.	Larger in size. A big monolithic OS (like Linux and Windows), on the other hand, provides far more services in the protected kernel space. Linux's kernel is over 2 million lines of code; Window's is far bigger. Monolithic kernels have been tremendously successful and do a yeoman's job of running the world's desktops and many embedded systems.
Age	Much younger.	Much older.
Bulk	Less bulky.	More bulky.
Drivers	The device drivers reside in the user space.	The device drivers reside in the kernel space.
Military use	Yes	No
Implementation	User services and kernel services are NOT implemented under same address space.	User services and kernel services are implemented under same address space.
Features	Not provided.	This kernel provides CPU scheduling, memory management, file management and other operating system functions through system calls.
Description	It is not a single static binary file. In Microkernels, the kernel is broken down into separate processes, known as servers.	It is a single static binary file. Monolithic kernel is a single large processes running entirely in a single address space.
Speed	Slower.	Faster.
Execution	Slow execution.	Fast execution.
Extensibility	Easily extensible.	Hard to extend.
Stability	If a service crashes, it has effects on the working of the OS.	If the service crashes, the whole system crashes.
	If a server crashes, the kernel can restart that component rather than having the entire system die or fall into a seriously degraded mode. Advocates of the monolithic kernel note that the microkernel is far from a panacea and that malicious attacks can still cripple the system.	
Security	Results provide very strong evidence that operating system structure has a strong effect on security. 96% of critical Linux exploits would not reach critical severity in a microkernel-based system, 57% would be reduced to low severity, the majority of which would be eliminated altogether if the system was based on a verified microkernel. Even without verification, a	

	microkernel-based design alone would completely prevent 29% of exploits. From the security point of view, the monolithic OS design is flawed and a root cause of the majority of compromises.	
Code	More code is required to write a microkernel.	Lesser code is required.
Adding new features	Does not require re-compilation.	Requires re-compilation.
Function calling	The communication in microkernels is done via message passing. The servers communicate through IPC (Interprocess Communication). Servers invoke "services" from each other by sending messages.	The kernel can invoke functions directly.
Threat Model	In contrast, in a microkernel-based system, the TCB of an application becomes highly dependent on the system services it uses.	In a monolithic OS, compromising one (kernel provided) service compromises the whole system, therefore the whole multi-million-SLOC kernel is in every application's TCB.
Benchmarking	The conservative criterion can be evaluated by benchmarks running on the same hardware platform under a monolithic and a microkernel-based operating system. This method measures not only the primary (direct) IPC costs but also the secondary costs induced by structuring software with the client/server paradigm and by using the IPC mechanism.	
Example	QNX, Symbian, L4Linux, etc.	Unix, Linux, BSDs, etc

A3:

Current operating systems have two characteristics that make them unreliable and insecure: They are huge and they have very poor fault isolation. The Linux kernel has more than 2.5 million lines of code; the Windows XP kernel is more than twice as large. One study of software reliability showed that code contains between six and 16 bugs per 1,000 lines of executable code.

OS Security.

OS security is traditionally characterized through the CIA group of three. These three points shape the reason for any safe OS. In the first place, Confidentiality guarantees that there is no revelation of information that has not been approved to any individual who isn't approved to get the data. Second, Integrity has the objective of ensuring the OS is free from ill-advised or unapproved OS changes, either equipment or programming based. In conclusion, Accessibility is basically portrayed as the OS's status for administration. In this manner OS security is condensed in these three. The security of the OS relies upon the conjunction of these three traits, necessitating that they should all exist together at the same time so as to protect security. Basically, OS security guarantees that the OS is doing what it should do.

OS Reliability

OS Reliability is a part of OS steadfastness; it is, notwithstanding, more complex than OS security. On the off chance that a OS is reliable, it must pursue that the same OS is solid. OS trustworthiness holds the traits of the CIA group of three of OS security, adding to its three additional traits: Safety, Maintainability, and Reliability. Security is depicted as the "nonattendance of calamitous results on the clients and nature." Maintainability essentially depicts the capacity of a OS to experience fixes, support, and other OS modifications, for example, fixes and refreshes. Reliability can be condensed as guaranteeing the "coherence of right administration"

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`Differences and Similarities

OS security and reliability are opposite sides of a similar coin; anyway they dwell in two "unique spaces of reasoning." On one hand, OS security guarantees that attacks don't break OS integrity and can recoup from an attack. Then again, OS reliability guarantees that the OS works accurately. Adjusting the well known programming designing maxim: OS security guarantees the OS is "making the best choice," and OS reliability guarantees the OS is "doing the thing right." Both security and reliability originate from a thought situated in the need to "portray accuracy." "The 'reliability of security' is regularly considered however the 'security of reliability' isn't frequently considered." Although both are basic to one another, one may get an inclination that a OS is secure on the off chance that it is dependable; and then again one may get the inclination that a OS is secure in the event that it is thusly, dependable. As we expressed already, blame aversion is a major piece of OS reliability, and here the two definitions are meeting. Security in OSs is planned to safeguard the OS against any dangers or programmers. For reliability, necessities are given to invigorate the plan of an item to support accessibility amidst equipment also, programming flaws. For security, prerequisites are given to invigorate the plan of an item to continue accessibility amidst security abnormalities. Both OS reliability and OS security are utilized as the premise of blame evacuation system.